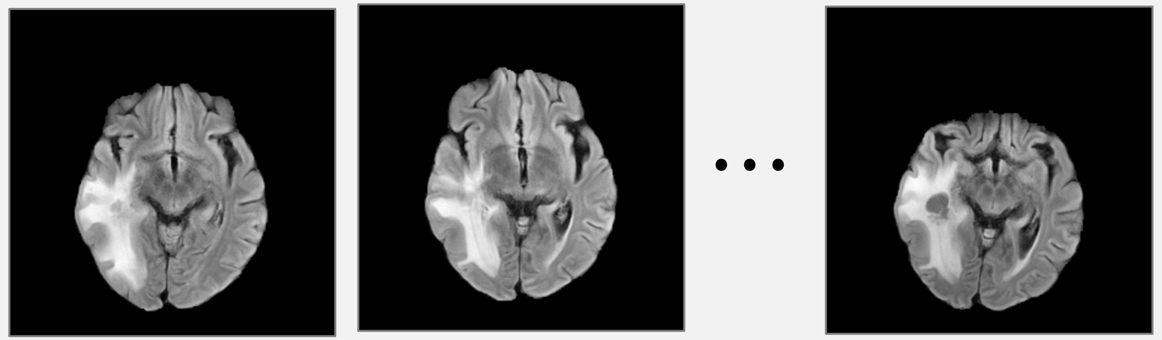
Recurrent metastatic lesion

Question: Which of the following best describes the temporal evolution of the left temporal lesion?



Options:

A. Progressive enlargement with a new necrotic component
B. Stable lesion with unchanged surrounding edema
C. Partial regression with decreased lesion enhancement
D. Complete resolution without residual edema

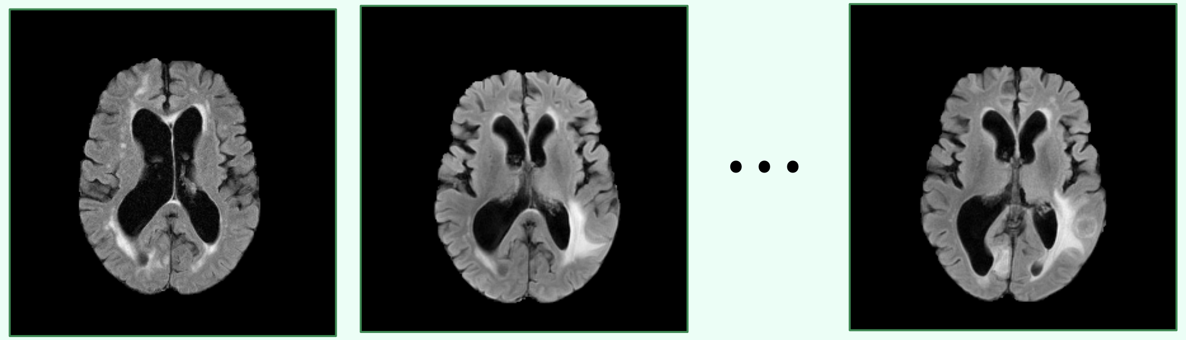
Steps:

1. Compare lesion extent and surrounding signal abnormality across serial MRIs.
2. Observe a reduction in lesion size and associated signal abnormality.
3. Note persistent but reduced surrounding abnormality without a new enlarging component.
4. Conclude that the lesion demonstrates partial regression rather than progression.

Answer: C

Glioblastoma multiforme (GBM)

Question: The progressive marginal enhancement around the resection cavity is concerning for recurrent tumor growth.



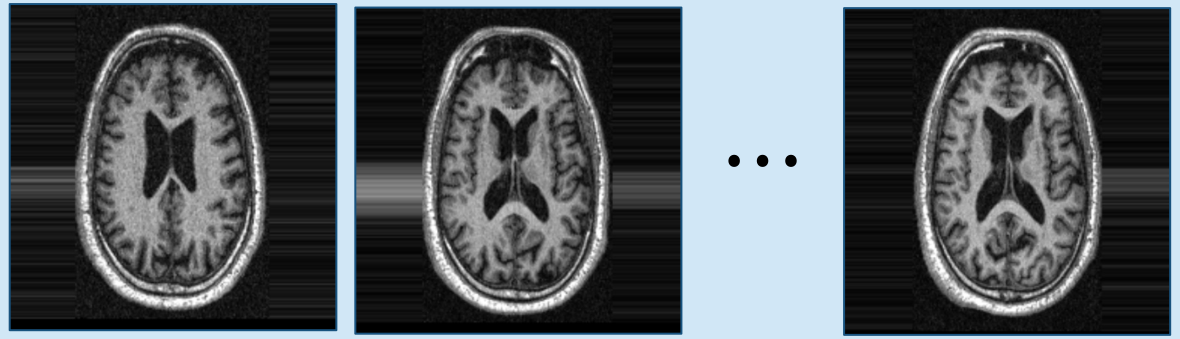
Steps:

1. Compare enhancement patterns across timepoints.
2. Observe progressive expansion of the enhancing margins.
3. Consider treatment-related change as an alternative explanation.
4. The interval increase is concerning for disease progression.

Answer: "True"

Aging / Alzheimer's progression

Question: How have cerebral volume and ventricular size changed from baseline to the latest follow-up MRI?



Steps:

1. The baseline MRI shows relatively preserved cerebral volume with mild ventricular prominence.
2. The intermediate scan shows increasing ventricular and sulcal prominence.
3. The latest scan demonstrates more pronounced cerebral volume loss and ventriculomegaly.
4. These findings indicate progressive cerebral atrophy, compatible with neurodegenerative progression.

Answer: Progressive cerebral volume loss with ventricular and sulcal enlargement is observed, compatible with neurodegenerative progression.